

Qatar-1b

R-0639

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_220921 · created 2026-07-25T18:36:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459843.8034 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.055
Transit depth (Rp/Rs)^2	1.1 +/- 1.14 [%]
Orbital Inclination (inc)	79.29 +/- 2.88
Airmass coefficient 1 (a1)	2.06 +/- 0.95
Airmass coefficient 2 (a2)	-0.49 +/- 0.38
Scatter in the residuals of the lightcurve fit is	44.11 %
Best Comparison Star	None
Optimal Aperture	8.28
Optimal Annulus	21.19
Transit Duration (day)	0.02 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.105 ± 0.055 vs 0.1463 (deviation 0.75σ)
Duration fit vs published	0.02 d vs 0.0692 d (deviation 1.89σ)
Mid-time O–C	-2.7 min
Depth SNR	1.9
Residual scatter	44.11 %

Light curve

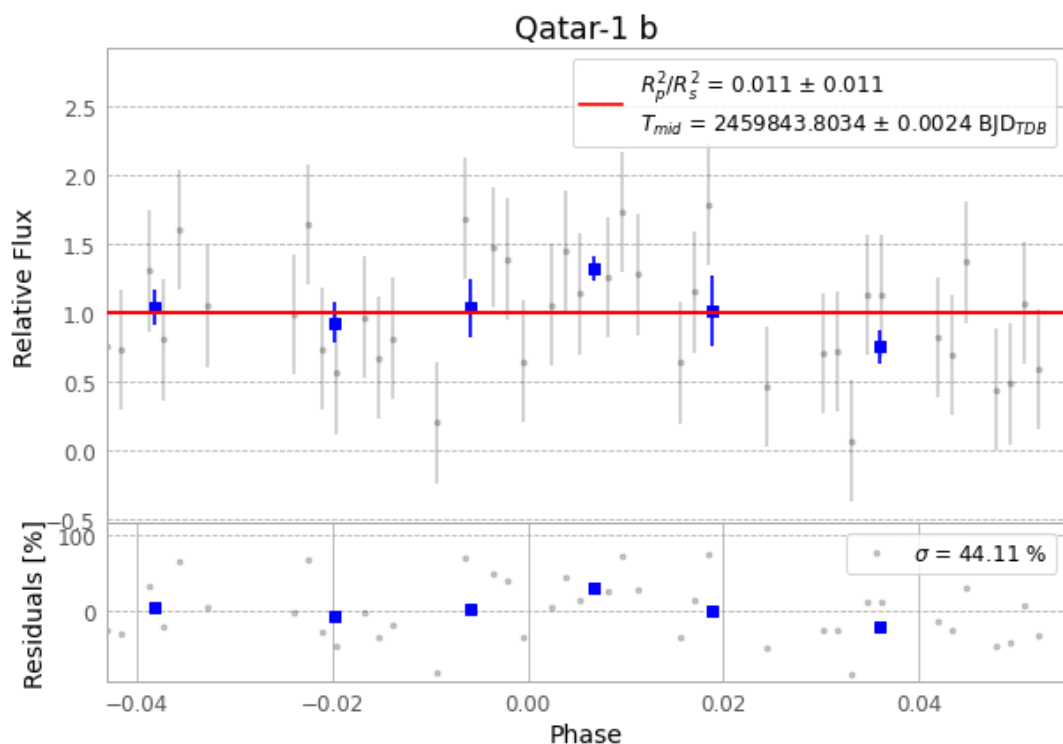


Figure 1 — FinalLightCurve_Qatar-1 b_2022-09-20.png (SHA-256 ee0ebc2e27ced842...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [393, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ce069c72eb5f4eec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

70 FITS frames (46 MB), Qatar-1220921053614.FITS → Qatar-1220921090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-220921.log (e6e6c7a3a6b7...), FinalLightCurve_Qatar-1 b_2022-09-20.png (ee0ebc2e27ce...), FinalLightCurve_Qatar-1 b_2022-09-20.csv (4394db74158d...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 18:36Z.